# 2024, Mar. 2024 Release LOC:Acute Pediatric
# Acute Kidney Injury

**Utilization Benchmarks**

**Length of Stay:** **3.7 d**
**Length of Stay Type:** **InterQual**
**Condition/Procedure:** **ACUTE KIDNEY INJURY**
**% Paid as Observation:** **27%**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

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## Overview

**Introduction:**
Acute kidney injury (AKI) is characterized by an abrupt deterioration in kidney function manifested by an increase in serum creatinine and reduced urine output. AKI is associated with significant morbidity and mortality in hospitalized patients. Early detection and treatment are essential to prevent adverse outcomes. AKI includes patients with chronic kidney disease who experience an acute deterioration.
Causes include (Ostermann et al., Kidney Int 2020, 98: 294-309; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019):

- Infection
- Inflammation
- Nephrotoxicity
- Obstruction
- Reduced perfusion

**Evaluation:**
Initial evaluation includes (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138):

- Identification of any nephrotoxic substance (e.g., medications, contrast medium, or toxins) potentially impairing renal function.
- Physical examination with assessment of intravascular volume status and potential signs of a systemic illness.
- Initial laboratory evaluation, including serum creatinine, blood urea nitrogen (BUN), complete blood count,

urinalysis. Additional evaluation may include a 24-hour urine test and examination of a urine sediment.
- Ultrasonography is recommended in all patients who present with acute kidney injury (AKI) to exclude obstructive uropathy and to measure renal size and cortical thickness.
- Renal biopsy is considered for patients with clinical or laboratory evidence of nephritis or unknown etiology of kidney injury.

**Treatment:**
Management of acute kidney injury (AKI) depends on the etiology, and can include one or more of the following:
- Fluid resuscitation and optimization of renal perfusion
- Discontinuation of nephrotoxic medications
- Correction of electrolyte imbalances

Dialysis is indicated for:
- Refractory hyperkalemia
- Volume overload
- Intractable acidosis
- Uremic encephalopathy
- Pericarditis
- Pleuritis
- Toxin removal

Treatable causes of AKI that should be identified and aggressively managed are (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138):
- Vasculitis
- Acute interstitial nephritis
- Acute glomerulonephritis
- Lupus nephritis
- Scleroderma
- Acute tubular necrosis
- Thrombotic microangiopathy

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kaddourah et al., N Engl J Med 2017, 376: 11-20; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

(Excludes PO medications unless noted)

**Select Day, One:**

- **Episode Day 1, One:**
  (Symptom or finding within 24h)
  - **OBSERVATION, Both:** [1]
    - Finding, ≥ **One:**
      - Urine output, ≥ **One:** [2]
        - ≤ 1 mL/kg/h
        - < 180 mL/$m^2$/24h
      - Creatinine, ≥ **One:**
        - Increased by ≥ 50%(0.50) within last 7d
        - Increased by ≥ 0.3 mg/dL(26.5 µmol/L) within last 2d
        - ≥ 1.5x baseline and chronic kidney disease
        - ≥ 1.5x upper limit of normal (ULN) [3, 4]
      - Glomerular filtration rate (GFR) > 25%(0.25) decrease from baseline [5]
    - Intervention, ≥ **One:**
      - Diuretic ≥ 2x/24h
      - IV fluid [6, 7]
      - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [8]
        - Diuretic (includes PO)
        - Nephrotoxic agent (includes PO) [9]
  - **ACUTE, One:** [10]
    - Intrinsic kidney disease or vasculitis, actual or suspected and, **All:** [11]
      - Finding, ≥ **One:**
        - Urine output, ≥ **One:** [2]
          - ≤ 1 mL/kg/h
          - < 180 mL/$m^2$/24h
        - Creatinine, **One:**
          - ≥ 2x baseline and chronic kidney disease [5]
          - ≥ 2x upper limit of normal (ULN)
        - Glomerular filtration rate (GFR) > 50%(0.50) decrease from baseline [5]
      - Diagnostic evaluation, ≥ **One:** [12]
        - Diagnosis established
        - Renal biomarkers, ≥ **One:**
          - Antinuclear antibody (ANA)
          - Antineutrophil cytoplasmic antibody (ANCA)
          - Anti-double stranded DNA antibody (Anti-dsDNA)
          - Anti-glomerular basement membrane (Anti-GBM)
        - Renal biopsy planned or performed
        - Renal ultrasound
      - Intervention, ≥ **One:**
        - Volume expander ≤ 2d [13, 14]
        - Diuretic ≥ 2x/24h, ≤ 3d
        - Immunosuppressant therapy ≤ 3d [15, 16]
        - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [8]
          - Diuretic (includes PO)
          - Nephrotoxic agent (includes PO) [9]
    - Acute kidney injury, hospital acquired and, **Both:** [17, 18]
      - Finding, ≥ **One:**
        - Urine output, ≥ **One:**
          - ≤ 1 mL/kg/h
          - < 180 mL/$m^2$/24h
        - Creatinine, **One:**

- ≥ 1.5x baseline and chronic kidney disease
- ≥ 1.5x ULN [3, 4]
- GFR > 25%(0.25) decrease from baseline [5]
- Dialysis initiated
- Intervention, ≥ **One:**
  - Volume expander ≤ 2d [13, 14]
  - Diuretic ≥ 2x/24h, ≤ 3d
  - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [8]
    - Diuretic (includes PO)
    - Nephrotoxic agent (includes PO) [9]
  - Dialysis (initial), **One:** [19]
    - Hemodialysis ≤ 7d since initiation
    - Peritoneal ≤ 7d since initiation
- **CRITICAL, ≥ One:** [20]
  - Acute kidney injury (AKI) and, **Both:**
    - Finding, ≥ **One:**
      - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
        - Loss of P wave
        - Multifocal PVCs
        - Neuromuscular deficit [21]
        - Widening QRS
      - Sodium, **One:**
        - < 120 mEq/L(120 mmol/L) and volume overload
        - > 158 mEq/L(158 mmol/L)
    - Intervention, ≥ **One:**
      - Calcium chloride
      - Calcium gluconate
      - Glucose 50%(0.50) with insulin [22]
      - Diuretic, continuous
      - Dialysis, **One:**
        - Hemodialysis ≤ 7d since initiation
        - Peritoneal dialysis ≤ 7d since initiation
      - IV fluid resuscitation ≤ 2d [23]
  - CRRT [24]
- **Episode Day 2, One:**
  - **OBSERVATION, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **Both:** [25]
      - Kidney function improving or within acceptable limits [26]
      - Urine output improving or within acceptable limits [26]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
      - Dialysis initiation (use **Acute Kidney Injury, Episode Day 2, ACUTE**) [27]
      - Acute kidney injury (AKI) and, **Both:** [17]
        - Finding, ≥ **One:**
          - Urine output, ≥ **One:**
            - ≤ 1 mL/kg/h
            - < 180 mL/$m^2$/24h
          - Creatinine, **One:**
            - > baseline and chronic kidney disease
            - > upper limit of normal (ULN) [3, 4]
        - Intervention, ≥ **One:**
          - Diuretic ≥ 2x/24h, ≤ 3d
          - IV fluid [6, 7]
          - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [8]

- Diuretic (includes PO)
- Nephrotoxic agent (includes PO) [9]

- **ACUTE, One:**
  - **Responder**, discharge expected today if clinically stable last 24h, **All:** [25]
    - Afebrile
    - Kidney function improving or within acceptable limits [26]
    - Complication or comorbidity, **One:**
      - No complication or active comorbidity relevant to this episode of care [28]
      - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits [26]
        - Outpatient dialysis regimen established
        - Pneumonia and, **Both:**
          - $O_2$ sat ≥ 94%(0.94) or within acceptable limits [26]
          - WBC, ≥ **One:** [29]
            - Age ≤ 12 mos and, **Both:**
              - WBC > 5,000/cu.mm($5x10^9$/L)
              - WBC < 17,500/cu.mm($17.5x10^9$/L)
            - Age > 1 to 5 yrs and, **Both:**
              - WBC > 6,000/cu.mm($6x10^9$/L)
              - WBC < 15,500/cu.mm($15.5x10^9$/L)
            - Age > 5 to 12 yrs and, **Both:**
              - WBC > 4,500/cu.mm($4.5x10^9$/L)
              - WBC < 13,500/cu.mm($13.5x10^9$/L)
            - Age > 12 to < 18 yrs and, **Both:**
              - WBC > 4,500/cu.mm($4.5x10^9$/L)
              - WBC < 12,000/cu.mm($12x10^9$/L)
            - Bands ≤ 10%(0.10)
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Acute kidney injury (AKI) and, **Both:** [17]
      - Finding, ≥ **One:**
        - Urine output, ≥ **One:**
          - ≤ 1 mL/kg/h
          - < 180 mL/$m^2$/24h
        - Creatinine, **One:**
          - ≥ 1.5x baseline and chronic kidney disease
          - ≥ 1.5x ULN [3, 4]
        - GFR > 25%(0.25) decrease from baseline [5]
        - Dialysis initiated
      - Intervention, ≥ **One:**
        - Volume expander ≤ 2d [13, 14]
        - Diuretic ≥ 2x/24h, ≤ 3d
        - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [8]
          - Diuretic (includes PO)
          - Nephrotoxic agent (includes PO) [9]
        - Dialysis (initial) and ≤ 7d since initiation [19]
        - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation [30]
        - Immunosuppressant therapy ≤ 3d [15, 16]
    - Blood glucose ≤ 24h, **All:** [31]
      - Finding, ≥ **One:**
        - > 250 mg/dL(13.9 mmol/L)
        - < 70 mg/dL(3.9 mmol/L)
      - Blood glucose monitoring at least 4x/24h
      - Medication adjustment and, ≥ **One:** [8, 31]

- ⊖ Insulin, **One:** (32)
  - − IV ≥ 2x/24h
  - − SC ≥ 4x/24h
- − Oral hypoglycemic agent
- ⊖ Pneumonia confirmed by imaging and, **Both:** (33)
  - ⊖ Finding, ≥ **One:**
    - − Temperature ≥ 100.4°F(38.0°C) (34)
    - − Mental status change (excludes coma, stupor, or somnolence) (35)
    - − $O_2$ sat ≤ 93%(0.93)
    - ⊖ WBC, **One:** (29)
      - ⊖ Age ≤ 12 mos and, **One:**
        - − WBC ≤ 5,000/cu.mm($5x10^9$/L)
        - − WBC ≥ 17,500/cu.mm($17.5x10^9$/L)
      - ⊖ Age > 1 to 5 yrs and, **One:**
        - − WBC ≤ 6,000/cu.mm($6x10^9$/L)
        - − WBC ≥ 15,500/cu.mm($15.5x10^9$/L)
      - ⊖ Age > 5 to 12 yrs and, **One:**
        - − WBC ≤ 4,500/cu.mm($4.5x10^9$/L)
        - − WBC ≥ 13,500/cu.mm($13.5x10^9$/L)
      - ⊖ Age > 12 to < 18 yrs and, **One:**
        - − WBC ≤ 4,500/cu.mm($4.5x10^9$/L)
        - − WBC ≥ 12,000/cu.mm($12x10^9$/L)
    - − Bands > 10%(0.10)
  - ⊖ Intervention, ≥ **One:** (36)
    - − Requiring supplemental oxygen (37)
    - − Anti-infective (includes PO)
    - − Respiratory interventions at least 6x/24h (38)
- − Post Critical care ≤ 24h

⊖ **CRITICAL, ≥ One:**
- ⊖ Acute kidney injury (AKI) and, **Both:**
  - ⊖ Finding, ≥ **One:**
    - ⊖ Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
      - − Loss of P wave
      - − Multifocal PVCs
      - − Neuromuscular deficit (21)
      - − Widening QRS
    - ⊖ Sodium, **One:**
      - − < 120 mEq/L(120 mmol/L) and volume overload
      - − > 158 mEq/L(158 mmol/L)
  - ⊖ Intervention, ≥ **One:**
    - − Calcium chloride
    - − Calcium gluconate
    - − Glucose 50%(0.50) with insulin (22)
    - − Diuretic, continuous
    - ⊖ Dialysis, **One:**
      - − Hemodialysis ≤ 7d since initiation
      - − Peritoneal dialysis ≤ 7d since initiation
    - − IV fluid resuscitation ≤ 2d (23)
- − CRRT (24)

⊖ **Episode Day 3, One:**
- ⊖ **OBSERVATION, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **Both:** (25)
    - − Kidney function improving or within acceptable limits (26)

- Urine output improving or within acceptable limits [26]

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
  - For acute kidney injury (AKI) use Episode Day 3 at a higher level of care
  - For a condition other than acute kidney injury (AKI) use appropriate subset (Episode Day 1) based on clinical findings
  - Refer for secondary review
- **ACUTE, One:**
  - **Responder**, discharge expected today if clinically stable last 24h, **All:** [25]
    - Afebrile
    - Kidney function improving or within acceptable limits [26]
    - Complication or comorbidity, **One:**
      - No complication or active comorbidity relevant to this episode of care [28]
      - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits [26]
        - Outpatient dialysis regimen established
        - Pneumonia and, **Both:**
          - $O_2$ sat ≥ 94%(0.94) or within acceptable limits [26]
          - WBC, ≥ **One:** [29]
            - Age ≤ 12 mos and, **Both:**
              - WBC > 5,000/cu.mm($5x10^9$/L)
              - WBC < 17,500/cu.mm($17.5x10^9$/L)
            - Age > 1 to 5 yrs and, **Both:**
              - WBC > 6,000/cu.mm($6x10^9$/L)
              - WBC < 15,500/cu.mm($15.5x10^9$/L)
            - Age > 5 to 12 yrs and, **Both:**
              - WBC > 4,500/cu.mm($4.5x10^9$/L)
              - WBC < 13,500/cu.mm($13.5x10^9$/L)
            - Age > 12 to < 18 yrs and, **Both:**
              - WBC > 4,500/cu.mm($4.5x10^9$/L)
              - WBC < 12,000/cu.mm($12x10^9$/L)
            - Bands ≤ 10%(0.10)
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Acute kidney injury (AKI) and, **Both:** [17]
      - Finding, ≥ **One:**
        - Urine output, ≥ **One:**
          - ≤ 1 mL/kg/h
          - < 180 mL/$m^2$/24h
        - Creatinine, **One:**
          - ≥ 1.5x baseline and chronic kidney disease
          - ≥ 1.5x ULN [3, 4]
        - GFR > 25%(0.25) decrease from baseline [5]
        - Dialysis initiated
      - Intervention, ≥ **One:**
        - Volume expander ≤ 2d [13, 14]
        - Diuretic ≥ 2x/24h, ≤ 3d
        - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [8]
          - Diuretic (includes PO)
          - Nephrotoxic agent (includes PO) [9]
        - Dialysis (initial) and ≤ 7d since initiation [19]
        - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation [30]
        - Immunosuppressant therapy ≤ 3d [15, 16]
    - Blood glucose ≤ 24h, **All:** [31]

- ⊖ Finding, ≥ **One:**
  - > 250 mg/dL(13.9 mmol/L)
  - < 70 mg/dL(3.9 mmol/L)
- Blood glucose monitoring at least 4x/24h
- ⊖ Medication adjustment and, ≥ **One:** (8, 31)
  - ⊖ Insulin, **One:** (32)
    - IV ≥ 2x/24h
    - SC ≥ 4x/24h
  - Oral hypoglycemic agent
- ⊖ Pneumonia confirmed by imaging and, **Both:** (33)
  - ⊖ Finding, ≥ **One:**
    - Temperature ≥ 100.4°F(38.0°C) (34)
    - Mental status change (excludes coma, stupor, or somnolence) (35)
    - $O_2$ sat ≤ 93%(0.93)
    - ⊖ WBC, **One:** (29)
      - ⊖ Age ≤ 12 mos and, **One:**
        - WBC ≤ 5,000/cu.mm($5x10^9$/L)
        - WBC ≥ 17,500/cu.mm($17.5x10^9$/L)
      - ⊖ Age > 1 to 5 yrs and, **One:**
        - WBC ≤ 6,000/cu.mm($6x10^9$/L)
        - WBC ≥ 15,500/cu.mm($15.5x10^9$/L)
      - ⊖ Age > 5 to 12 yrs and, **One:**
        - WBC ≤ 4,500/cu.mm($4.5x10^9$/L)
        - WBC ≥ 13,500/cu.mm($13.5x10^9$/L)
      - ⊖ Age > 12 to < 18 yrs and, **One:**
        - WBC ≤ 4,500/cu.mm($4.5x10^9$/L)
        - WBC ≥ 12,000/cu.mm($12x10^9$/L)
    - Bands > 10%(0.10)
  - ⊖ Intervention, ≥ **One:** (36)
    - Requiring supplemental oxygen (37)
    - Anti-infective (includes PO)
    - Respiratory interventions at least 6x/24h (38)
- Post Critical care ≤ 24h

⊖ **CRITICAL,** ≥ **One:**
- ⊖ Acute kidney injury (AKI) and, **Both:**
  - ⊖ Finding, ≥ **One:**
    - ⊖ Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
      - Loss of P wave
      - Multifocal PVCs
      - Neuromuscular deficit (21)
      - Widening QRS
    - ⊖ Sodium, **One:**
      - < 120 mEq/L(120 mmol/L) and volume overload
      - > 158 mEq/L(158 mmol/L)
  - ⊖ Intervention, ≥ **One:**
    - Calcium chloride
    - Calcium gluconate
    - Glucose 50%(0.50) with insulin (22)
    - Diuretic, continuous
    - ⊖ Dialysis, **One:**
      - Hemodialysis ≤ 7d since initiation
      - Peritoneal dialysis ≤ 7d since initiation
    - IV fluid resuscitation ≤ 2d (23)

- CRRT [24]
- **Episode Day 4-7, One:**
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 24h, **All:** [25]
      - Afebrile
      - Kidney function improving or within acceptable limits [26]
      - Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [28]
        - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits [26]
          - Outpatient dialysis regimen established
          - Pneumonia and, **Both:**
            - $O_2$ sat ≥ 94%(0.94) or within acceptable limits [26]
            - WBC, ≥ **One:** [29]
              - Age ≤ 12 mos and, **Both:**
                - WBC > 5,000/cu.mm($5x10^9$/L)
                - WBC < 17,500/cu.mm($17.5x10^9$/L)
              - Age > 1 to 5 yrs and, **Both:**
                - WBC > 6,000/cu.mm($6x10^9$/L)
                - WBC < 15,500/cu.mm($15.5x10^9$/L)
              - Age > 5 to 12 yrs and, **Both:**
                - WBC > 4,500/cu.mm($4.5x10^9$/L)
                - WBC < 13,500/cu.mm($13.5x10^9$/L)
              - Age > 12 to < 18 yrs and, **Both:**
                - WBC > 4,500/cu.mm($4.5x10^9$/L)
                - WBC < 12,000/cu.mm($12x10^9$/L)
              - Bands ≤ 10%(0.10)
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Acute kidney injury (AKI) and, **Both:** [17]
        - Finding, ≥ **One:**
          - Urine output, ≥ **One:**
            - ≤ 1 mL/kg/h
            - < 180 mL/$m^2$/24h
          - Creatinine, **One:**
            - ≥ 1.5x baseline and chronic kidney disease
            - ≥ 1.5x ULN [3, 4]
          - GFR > 25%(0.25) decrease from baseline [5]
          - Dialysis initiated
        - Intervention, ≥ **One:**
          - Volume expander ≤ 2d [13, 14]
          - Diuretic ≥ 2x/24h, ≤ 3d
          - Medication adjustment or discontinuation ≤ 3d and, ≥ **One:** [8]
            - Diuretic (includes PO)
            - Nephrotoxic agent (includes PO) [9]
          - Dialysis (initial) and ≤ 7d since initiation [19]
          - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation [30]
          - Immunosuppressant therapy ≤ 3d [15, 16]
      - Blood glucose ≤ 24h, **All:** [31]
        - Finding, ≥ **One:**
          - > 250 mg/dL(13.9 mmol/L)
          - < 70 mg/dL(3.9 mmol/L)
        - Blood glucose monitoring at least 4x/24h
        - Medication adjustment and, ≥ **One:** [8, 31]

            - Insulin, **One:** (32)
                - IV ≥ 2x/24h
                - SC ≥ 4x/24h
            - Oral hypoglycemic agent
        - Pneumonia confirmed by imaging and, **Both:** (33)
            - Finding, ≥ **One:**
                - Temperature ≥ 100.4°F(38.0°C) (34)
                - Mental status change (excludes coma, stupor, or somnolence) (35)
                - $O_2$ sat ≤ 93%(0.93)
                - WBC, **One:** (29)
                    - Age ≤ 12 mos and, **One:**
                        - WBC ≤ 5,000/cu.mm($5x10^9$/L)
                        - WBC ≥ 17,500/cu.mm($17.5x10^9$/L)
                    - Age > 1 to 5 yrs and, **One:**
                        - WBC ≤ 6,000/cu.mm($6x10^9$/L)
                        - WBC ≥ 15,500/cu.mm($15.5x10^9$/L)
                    - Age > 5 to 12 yrs and, **One:**
                        - WBC ≤ 4,500/cu.mm($4.5x10^9$/L)
                        - WBC ≥ 13,500/cu.mm($13.5x10^9$/L)
                    - Age > 12 to < 18 yrs and, **One:**
                        - WBC ≤ 4,500/cu.mm($4.5x10^9$/L)
                        - WBC ≥ 12,000/cu.mm($12x10^9$/L)
                - Bands > 10%(0.10)
            - Intervention, ≥ **One:** (36)
                - Requiring supplemental oxygen (37)
                - Anti-infective (includes PO)
                - Respiratory interventions at least 6x/24h (38)
        - Post Critical care ≤ 24h
    - **CRITICAL, ≥ One:**
        - Acute kidney injury (AKI) and, **Both:**
            - Finding, ≥ **One:**
                - Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One:**
                    - Loss of P wave
                    - Multifocal PVCs
                    - Neuromuscular deficit (21)
                    - Widening QRS
                - Sodium, **One:**
                    - < 120 mEq/L(120 mmol/L) and volume overload
                    - > 158 mEq/L(158 mmol/L)
            - Intervention, ≥ **One:**
                - Calcium chloride
                - Calcium gluconate
                - Glucose 50%(0.50) with insulin (22)
                - Diuretic, continuous
                - Dialysis, **One:**
                    - Hemodialysis ≤ 7d since initiation
                    - Peritoneal dialysis ≤ 7d since initiation
                - IV fluid resuscitation ≤ 2d (23)
        - CRRT (24)
- **Episode Day 8, One:**
    - **ACUTE, One:**
        - **Responder**, discharge expected today if clinically stable last 24h, **All:** (25)
            - Afebrile

- Kidney function improving or within acceptable limits [26]
- ⊖ Complication or comorbidity, **One:**
  - No complication or active comorbidity relevant to this episode of care [28]
  - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
    - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits [26]
    - Outpatient dialysis regimen established
    - ⊖ Pneumonia and, **Both:**
      - $O_2$ sat ≥ 94%(0.94) or within acceptable limits [26]
      - ⊖ WBC, ≥ **One:** [29]
        - ⊖ Age ≤ 12 mos and, **Both:**
          - WBC > 5,000/cu.mm($5x10^9$/L)
          - WBC < 17,500/cu.mm($17.5x10^9$/L)
        - ⊖ Age > 1 to 5 yrs and, **Both:**
          - WBC > 6,000/cu.mm($6x10^9$/L)
          - WBC < 15,500/cu.mm($15.5x10^9$/L)
        - ⊖ Age > 5 to 12 yrs and, **Both:**
          - WBC > 4,500/cu.mm($4.5x10^9$/L)
          - WBC < 13,500/cu.mm($13.5x10^9$/L)
        - ⊖ Age > 12 to < 18 yrs and, **Both:**
          - WBC > 4,500/cu.mm($4.5x10^9$/L)
          - WBC < 12,000/cu.mm($12x10^9$/L)
        - Bands ≤ 10%(0.10)

- ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
  - **Use Extended Stay subset**
  - Refer for secondary review

⊖ **CRITICAL, One:**

- ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
  - **Use Extended Stay subset**
  - Refer for secondary review

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**Discharge, Level of Care, One:** [39]

- ⊖ **HOME, Both:**
  - ⊖ Level of care appropriateness, **Both:**
    - Home environment safe and accessible [40]
    - Patient or caregiver demonstrates ability to manage care
  - ⊖ AKI Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [41]
    - ⊖ Education on the importance of acute kidney injury care, **Both:**
      - Monitor daily weight
      - Monitor daily sodium intake
    - Medication reconciliation [42]
    - Follow-up care planned [43]
    - Identify and address transportation needs
- ⊖ **HOME CARE, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Home environment safe and accessible [40]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [44, 45]
    - ⊖ Skilled services, ≥ **One:** [46]
      - Skilled nursing [47]
      - Skilled therapy, PT, OT, or ST

- Behavioral health [48]
- ⊖ AKI Discharge planning, **All:**
  - Provide comprehensive written discharge and teaching instructions [41]
  - ⊖ Education on the importance of acute kidney injury care, **Both:**
    - Monitor weight daily
    - Monitor daily sodium intake
  - Medication reconciliation [42]
  - Follow-up care planned [43]
  - Identify and address transportation needs
- ⊖ **BEHAVIORAL HEALTH, Both:**
  - ⊖ Level of care appropriateness, **One:**
    - Support systems available [49]
  - ⊖ Skilled treatment, ≥ **One:**
    - Clinical assessment [50]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [51]
- ⊖ **SKILLED MEDICAL OR THERAPY, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [47]
      - ⊖ Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [52, 53]
        - Functional impairments requiring at least supervision [54]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation [42]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **SUBACUTE MEDICAL OR THERAPY, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [47]
      - ⊖ Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [52, 53]
        - ≥ 2 functional impairments requiring at least minimum assistance [54]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation [42]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **SUBACUTE REHAB, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, **All:**
      - Goal directed therapy with rehabilitation potential [52, 53]
      - ≥ 2 functional impairments requiring at least minimum assistance [54]

      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [42]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [52, 53]
      - ≥ 2 functional impairments requiring at least minimum assistance [54]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [55]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [42]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:** [56]
      - Primary condition or illness and treatment of active comorbid conditions [57]
      - Ventilator dependent ≥ 6h/d and weaning planned [58, 59]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [42]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation

## Notes:

**1:**

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
**Note**: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

**2:**

**Instruction:** Measuring urine output in newborns, infants, or patients who are unable to use a bedpan or other measuring device can be done via straight catheterization, or alternatively, by weighing diapers. Weight in grams can be translated to milliliters (1 gram = 1 mL) (Fuhrman, Pediatr Crit Care Med 2019, 20: 381-2).

**3:**

Creatinine elevation reflects renal function impairment. The following values are equivalent to 1.5x upper limit of normal (ULN) by age: (McMillan et al., The Harriet Lane Handbook. 2008)
**Age/Creatinine**

- 3 days to 1 year: ≥ 0.6 mg/dL(53 µmol/L)
- > 1 year to 3 years: ≥ 0.9 mg/dL(80 µmol/L)
- > 3 years to 11 years: ≥ 1.1 mg/dL(97 µmol/L)
- 12 years to < 18 years: ≥ 1.5 mg/dL(133 µmol/L)

**4:**

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

**5:**

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

**6:**

To allow for the advancement of feedings as intravenous (IV) fluids are weaned, the IV fluid rates are ≥ 50%(0.50) of the maintenance fluid requirement. These rates can be calculated using the following (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020):
Weight ≤ 10 *kg* = ≥ 50 *mL/kg*/24h

- Hourly rate = 2.08 (*mL/h*) x kg

Weight > 10-25 *kg* = ≥ 40 *mL/kg*/24h

- Hourly rate = 1.67 (*mL/h*) x kg

Weight > 25-60 *kg* = ≥ 30 *mL/kg*/24h

- Hourly rate = 1.25 (*mL/h*) x kg

Example: An 11 *kg* child is required to receive a minimum of 40 *mL/kg*/24h to meet this criteria point. Multiply 1.67 (*mL*/h) x 11 (child's weight in *kg*) = 18.37 *mL*/h. This is the minimum fluid replacement rate for this child.

**7:**

**Instruction:** This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

**8:**

**Instruction:** For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

**9:**

Nephrotoxic agents include nonsteroidal anti-inflammatory drugs (NSAIDs), aminoglycosides, amphotericin B, high doses of loop diuretics, and anti-viral agents or other drugs (e.g., acyclovir and foscarnet) (Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

**10:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
**Note**: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

**11:**

Intrinsic kidney diseases result from damage to the kidney causing acute loss of kidney function, including acute interstitial nephritis, acute glomerulonephritis, lupus nephritis, scleroderma, acute tubular necrosis, and thrombotic microangiopathy. This excludes pre-renal azotemia.

**12:**

In patients with acute kidney injury (AKI), the diagnostic evaluation seeks to identify the etiology and to determine appropriate treatment. The causes of kidney disease may be classified into 3 broad categories: pre-renal, renal, and post-renal. Prerenal causes are often reversible and is usually caused by renal hypoperfusion due to hypovolemia. Intrinsic renal disease is accompanied by pathologic changes in the kidney (i.e., vasculature, glomeruli, tubules, interstitium). Diagnostic tests to identify the cause of the kidney dysfunction may include examination of urinary sediment, serum renal biomarkers, renal ultrasound, or renal biopsy. Prior to the initiation of Immunosuppressants for systemic diseases, which adversely impacts kidney function, a renal biopsy may be necessary. Postrenal disease results from obstruction. Early identification and relief of obstruction is essential for improved patient outcomes.

**13:**

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.
**Instruction:** In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

**14:**

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Pistolesi et al., J Nephrol 2018, 31: 797-812; Joannidis et al., Intensive Care Med 2017, 43: 730-49; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

**15:**

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

**16:**
**Instruction:** If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

**17:**
Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.
AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.
AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

**18:**
Hospital acquired refers to a condition that developed during the patient's current hospitalization. Criteria for hospital acquired conditions cannot be applied to patients who either have been discharged to home or transferred from a post-acute facility.

**19:**
Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

**20:**
Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
**Note**: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

**21:**
Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

**22:**
Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

**23:**
Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60

milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

**24:**

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.
These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

**25:**

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

**26:**

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

**27:**

**Instruction:** To conduct a review for patients with acute kidney injury at the Observation level of care who require initiation of dialysis, refer to the Acute level of care on Episode Day 2.

**28:**

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

**29:**

Reference for pediatric WBC range: (Dellinger et al., Crit Care Med 2013, 41: 580-637).

**30:**

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

**31:**

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

**32:**

**Instruction**: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

**33:**

**Instruction:** This criterion includes aspiration pneumonitis. Aspiration is marked by the abnormal movement of oropharyngeal or gastric contents into the lower respiratory tract. Aspiration pneumonitis occurs when acidic gastric contents cause inflammation of the lung parenchyma. In the early stages, aspiration pneumonitis is not considered an infectious process as gastric acid prevents microbial growth. Anti-infective therapy should be considered if symptoms (e.g., shortness of breath, wheezing, hypoxemia) fail to resolve within 48 hours (Neill and Dean, Curr Opin Infect Dis 2019, 32: 152-7).

**34:**

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

**35:**

For patients at the Acute level of care, mental status change may include disorientation, agitation, increasing irritability, listlessness, or lethargy. The inability to soothe an infant, abnormal cry (e.g., weak, high-pitched), or decreased social interaction with parents or caregivers in children 2 years old or younger, may also qualify as a mental status change (Keith Kleinman, In: Harriet Lane Handbook, 22nd edn. 2021).
**Instruction**: Patients with acute coma, stupor, or somnolence should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma or stupor.

**36:**

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

**37:**

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) $FIO_2$] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage ($FIO_2$) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of $FIO_2$ optimally delivered:
**Low-flow systems**

- Nasal cannula / Nasopharyngeal catheters 24-35%(*0.24-0.35*) $FIO_2$

**Reservoir systems** (For neonates, infants, and young children, the maximum $FIO_2$ achievable is not well documented)

- Simple oxygen masks 35-50%(*0.35-0.50*) $FIO_2$

- Partial-rebreathing masks 40-60%(*0.40-0.60*) $FIO_2$
- Non-rebreathing masks > 60%(*0.60*) $FIO_2$

**High-flow systems**

- Air-entrainment masks / nebulizers 24-40%(*0.24-0.40*) $FIO_2$
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(*0.21-1.0*) $FIO_2$

**Enclosure systems**

- Oxygen hoods / tents 21-100%(*0.21-1.0*) $FIO_2$

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

**38:**

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

**39:**

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

**40:**

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

**41:**

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

**42:**

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

**43:**

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

**44:**

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

**45:**

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

**46:**

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

**47:**

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

**48:**

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.
**Evaluation and Treatment**
Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

**49:**

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

**50:**

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems are identified. The clinical assessment includes:

- History and physical exam (vital signs, height and weight)
- Pain assessment includes: cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

**51:**

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use

Disorders products.

**52:**

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

**53:**

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

**54:**

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

**55:**

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

**56:**

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.

**57:**

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

**58:**

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

**59:**

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness